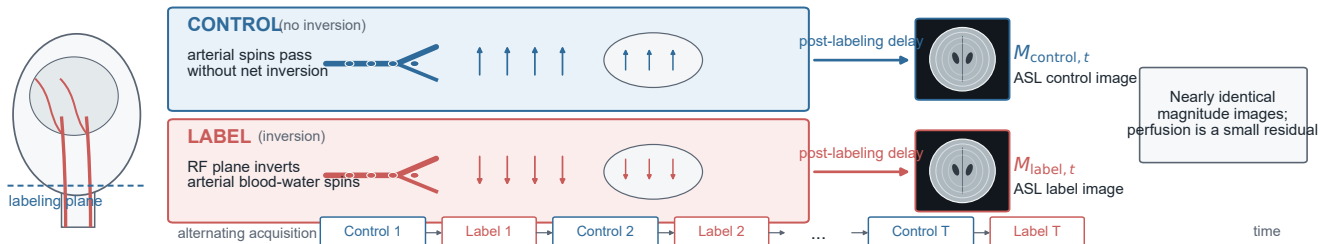


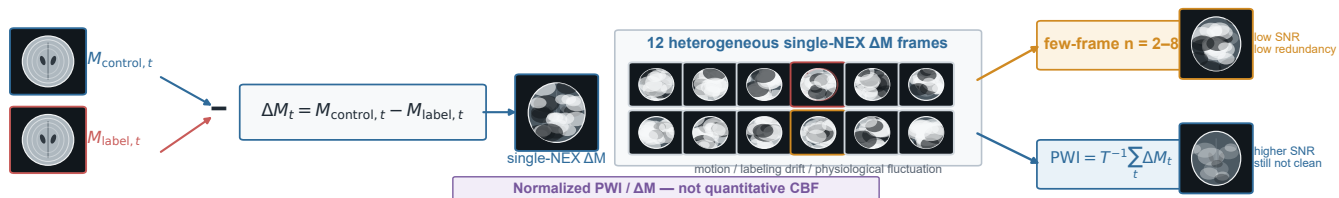
From arterial spin labeling to PWI: signal formation and project challenges

a ASL acquisition: endogenous arterial-water tracer

7 T • single PLD • 12 NEX

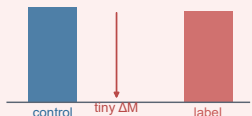


b Pairwise subtraction and repeated averaging



c Challenges addressed in this project

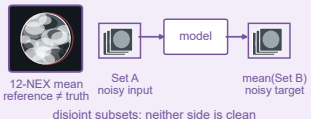
1 Weak subtraction signal
 $\Delta M \ll M_0$; subtraction exposes thermal and physiological noise.



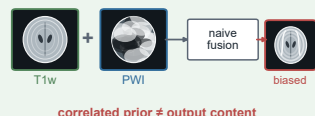
2 Scan time vs reliability
 More NEX improves averaging but lengthens the 7 T examination; one bad frame matters more.



3 No clean ground truth
 Structured motion, physiology and labeling drift persist across repetitions and their mean.



4 Auxiliary-anatomy bias
 T1w is sharper but is not perfusion; unrestricted fusion can transfer unsupported anatomical edges.



Design objective: robust few-frame reconstruction + self-supervision without clean targets + anatomical guidance without direct T1 content transfer